

[Home](#) / [Archives](#) / [Vol. 1 No. 1 \(2023\): Journal of Engineering and Computational Intelligence Review](#) / [Articles](#)

Real-Time Threat Intelligence Correlation and Triage for Reducing Security Analyst Burnout

Akib Rahman

Master of Information Systems Technologies (Information Assurance and Web Design), Wilmington University, New Castle, Delaware, USA

Author

Sharmin Sultana

Master of Information Systems Technologies (Information Assurance and Web Design), Wilmington University, New Castle, Delaware, USA

Author

Keywords: Cyber Threat Intelligence, Real-Time Correlation, Triage Automation, Graph Neural Networks, Reinforcement Learning, Analyst Burnout, Security Operations Center, Threat Prioritization, Machine Learning Security.

Abstract

Security Operations Centers (SOCs) face a critical crisis as cybersecurity analysts suffer from overwhelming burnout, with over 60% reporting exhaustion due to the manual processing of thousands of daily alerts. This chronic stress leads to decision fatigue, increased oversight of genuine threats, and compromised organizational security. To address this, we present AutoTI-Triage, an autonomous system for real-time threat intelligence correlation and triage designed to alleviate cognitive load and augment human decision-making.

AutoTI-Triage employs a hybrid architecture combining Graph Neural Networks (GNNs) and Reinforcement Learning (RL). The GNN component constructs dynamic threat graphs to map complex relationships between threat actors, indicators of compromise (IOCs), and assets, revealing hidden attack patterns. Concurrently, the RL agent learns optimal, adaptive triage policies from analyst feedback and incident outcomes, continuously refining prioritization accuracy. We validate our system using a comprehensive dataset of 1.2 million threat intelligence events from sources like AlienVault OTX and MISP, representing the largest public benchmark for TI correlation. Quantitative

evaluation demonstrates a 0.92 F1-score for threat classification, a 65% reduction in Mean Time to Resolution (MTTR), and significant gains in analyst productivity. By automating complex correlation and enabling adaptive prioritization, AutoTI-Triage offers a scalable solution to combat analyst burnout while enhancing the efficacy of modern security operations.

REFERENCES

- [1] D. Reinsel, J. Gantz, and J. Rydning, *The Digitization of the World: From Edge to Core*. IDC, 2018.
- [2] S. Brown, J. Harris, and E. Hutchins, "Threat intelligence platforms: The next generation," Gartner Research, 2019.
- [3] W. Tounsi and H. Rais, "A survey on technical threat intelligence in the age of sophisticated cyber attacks," *Comput. Security*, vol. 72, pp. 212–233, 2018.
- [4] A. B. Bakker and E. Demerouti, "Job demands–resources theory: Taking stock and looking forward," *J. Occup. Health Psychol.*, vol. 22, no. 3, pp. 273–285, 2017.
- [5] B. E. Strom *et al.*, *MITRE ATT&CK: Design and Philosophy*. MITRE Corp., 2018.
- [6] R. A. Martin, "Managing vulnerabilities in networked systems," *Computer*, vol. 34, no. 11, pp. 32–38, 2001.
- [7] W. U. Hassan *et al.*, "NoDoze: Combatting alert fatigue with automated analytics across SOC workflows," in *Proc. ACM SIGSAC CCS*, 2019, pp. 253–270.
- [8] Verizon, *Data Breach Investigations Report 2023*. Verizon Business, 2023.
- [9] S. C. Sundaramurthy *et al.*, "Turning contradictions into innovations: An ethnographic study of a security operations center," in *Proc. CHI*, 2016, pp. 4567–4578.
- [10] S. Brown, J. Harris, and E. Hutchins, "Threat intelligence platforms: The next generation," Gartner Research, 2019.
- [11] S. Qamar *et al.*, "Threat intelligence platforms: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 3289–3315, 2017.
- [12] T. D. Wagner *et al.*, "The correlation of cyber threat intelligence: A review," *Comput. Security*, vol. 88, 2019.
- [13] C. Sauerwein *et al.*, "Threat intelligence sharing platforms: An empirical study," *Comput. Security*, vol. 102, 2021.
- [14] S. Samtani, R. Chinn, and H. Chen, "Exploring emerging hacker assets and key hackers in dark web forums," *Decis. Support Syst.*, vol. 135, 2020.

- [15] IBM Security, *Cost of a Data Breach Report 2023*. IBM Corp., 2023.
- [16] ISC², "Cybersecurity workforce study 2022," 2022.
- [17] Cybersecurity Ventures, "Cybersecurity workforce shortage report 2023," 2023.
- [18] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. ICLR*, 2017.
- [19] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.
- [20] C. Sauerwein *et al.*, "A systematic literature review of threat intelligence sharing," *J. Cybersecurity*, vol. 5, no. 1, 2019.
- [21] C. Maslach and S. E. Jackson, "The measurement of experienced burnout," *J. Organ. Behav.*, vol. 2, no. 2, pp. 99–113, 1981.
- [22] F. B. Kokulu *et al.*, "The human factor in security operations," in *Proc. 28th USENIX Security Symp.*, 2019, pp. 1385–1402.
- [23] L. Chen, Y. Zhang, and X. Wang, "Measuring analyst workload in modern SOC's," *IEEE Trans. Dependable Secure Comput.*, vol. 17, no. 4, pp. 789–802, 2020.
- [24] J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, 1960.
- [25] OASIS, "STIX™ Version 2.0," 2017.
- [26] K. Rieck, P. Trinius, C. Willems, and T. Holz, "Automatic analysis of malware behavior using machine learning," *J. Comput. Security*, vol. 19, no. 4, pp. 639–668, 2011.
- [27] W. L. Hamilton, *Graph Representation Learning*. Morgan & Claypool, 2020.
- [28] J. Zhao *et al.*, "TIMiner: Automatically extracting and analyzing categorized cyber threat intelligence from social data," *Comput. Security*, vol. 95, 2020.
- [29] S. Bhatt, P. K. Manadhata, and L. Zomlot, "The operational role of Security Information and Event Management systems," *IEEE Secur. Privacy*, vol. 12, no. 5, pp. 35–41, 2014.
- [30] D. Swift, *Successful SIEM and Log Management Strategies*. SANS Institute, 2010.
- [31] G. Apruzzese, M. Andreolini, M. Marchetti, A. Venturi, and M. Colajanni, "Identifying and tracking malicious cyber campaigns via attacker profiling: A deep learning approach," *IEEE Trans. Inf. Forensics Secur.*, vol. 17, pp. 2313–2328, 2022.

- [32] D. Arp, E. Quiring, K. Rieck, and C. Wressnegger, "Dos and don'ts of machine learning in computer security," in *Proc. 31st USENIX Security Symp.*, 2022, pp. 3971–3988.
- [33] T. Nguyen and V. J. Reddi, "Deep reinforcement learning for cyber security," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 7, pp. 2865–2879, 2021.
- [34] Y. Li, Q. Liu, and Z. Wang, "Reinforcement learning for alert prioritization in SOCs," *IEEE Trans. Inf. Forensics Secur.*, vol. 17, pp. 2987–3001, 2022.
- [35] H. Chen, J. Liu, Y. Zhang, and X. Li, "Hybrid GNN-RL architecture for adaptive threat triage," in *Proc. 32nd USENIX Security Symp.*, 2023, pp. 1123–1140.
- [36] M. Vielberth *et al.*, "Security operations center: A systematic literature review," *Comput. Security*, vol. 97, 2020.
- Ponemon Institute, *The State of Cybersecurity Analyst Burnout 2022*. Ponemon Institute LLC, 2022.
- [37] M. Husák *et al.*, "SoK: The impact of alerting on security operations centers," in *IEEE Symp. Security Privacy (SP)*, 2022, pp. 105–122.
- [38] P. Jacobs, D. Williams, and J. Smith, "Automation and analyst burnout in SOC environments," *J. Cybersecurity Privacy*, vol. 1, no. 3, pp. 456–478, 2021.
- [39] C. Maslach, S. E. Jackson, and M. P. Leiter, *Maslach Burnout Inventory Manual*, 3rd ed. Consulting Psychologists Press, 1996.
- [40] C. Maslach, W. B. Schaufeli, and M. P. Leiter, "Job burnout," *Annu. Rev. Psychol.*, vol. 52, no. 1, pp. 397–422, 2001.
- [41] E. Demerouti, A. B. Bakker, F. Nachreiner, and W. B. Schaufeli, "The job demands resources model of burnout," *J. Appl. Psychol.*, vol. 86, no. 3, pp. 499–512, 2001.
- [42] C. Islam, M. A. Babar, and S. Nepal, "A survey of machine learning techniques applied to Security Operations Centers," *ACM Comput. Surveys*, vol. 52, no. 4, Art. 78, 2019.
- [43] A. D'Amico, K. Buchanan, J. Goodall, and D. Tesone, "Critical incident analysis using visualization and human factors," in *Proc. Hum. Factors Ergonomics Soc. Annu. Meeting*, vol. 49, no. 3, pp. 456–460, 2005.
- [44] P. Velickovic *et al.*, "Deep Graph Infomax," in *Proc. ICLR*, 2019.
- [45] J. Kreps, S. Narkhede, and J. Rao, "Kafka: A distributed messaging system for log processing," in *Proc. NetDB Workshop*, 2011.

- [46] OASIS, "STIX™ Version 2.1," 2021.
- [47] J. Barnes, "Data quality considerations for cyber threat intelligence," *J. Cybersecurity*, vol. 6, no. 1, 2020.
- [48] C. Wagner *et al.*, "MISP: The design and implementation of a collaborative threat intelligence sharing platform," in *Proc. ACM Workshop ISC*, 2016, pp. 49–56.
- [49] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
- [50] P. Veličković *et al.*, "Graph Attention Networks," in *Proc. ICLR*, 2018.
- [51] P. Bojanowski, E. Grave, A. Joulin, and T. Mikolov, "Enriching word vectors with subword information," *Trans. Assoc. Comput. Linguistics*, vol. 5, pp. 135–146, 2017.
- [52] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.
- [53] M. Fey and J. E. Lenssen, "Fast graph representation learning with PyTorch Geometric," *arXiv preprint arXiv:1903.02428*, 2019.
- [54] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2015.
- [55] M. Asif and A. Shaheen, "Creating a high-performance workplace by the determination of importance of job satisfaction, employee engagement, and leadership," *Journal of Business Insight and Innovation*, vol. 1, no. 2, pp. 9–15, 2022.
- [56] N. Shahid, M. Asif, and A. Pasha, "Effect of internet addiction on school going children," *Inverge Journal of Social Sciences*, vol. 1, no. 1, pp. 12–47, 2022, doi: 10.63544/ijss.v1i1.3.
- [57] H. A. Usama, M. Riaz, A. Khan, N. Begum, M. Asif, and M. Hamza, "Prohibition of alcohol in Quran and Bible (A research and analytical review)," *PalArch's Journal of Archaeology of Egypt/Egyptology*, vol. 19, no. 4, pp. 1202–1211, 2022.
- [58] S. H. Alizai, M. Asif, and Z. K. Rind, "Relevance of motivational theories and firm health," *International Journal of Management*, vol. 12, no. 3, pp. 1130–1137, 2021.
- [59] M. Asif, "Contingent effect of conflict management towards psychological capital and employees' engagement in financial sector of Islamabad," Ph.D. dissertation, Preston University, 2021, doi: 10.13140/RG.2.2.17616.79360.
- [60] Aurangzeb, M. Asif, and M. K. Amin, "Resources management and SME's performance," *Humanities & Social Sciences Reviews*, vol. 9, no. 3, pp. 679–689, 2021, doi: 10.18510/hssr.2021.9367.
- [61] D. Aurangzeb and M. Asif, "Role of leadership in digital transformation: A case of Pakistani

SMEs," in *Proc. Fourth Int. Conf. Emerging Trends in Engineering*, 2021.

[62] Aurangzeb, T. Mushtaque, M. N. Tunio, Z. Rehman, and M. Asif, "Influence of administrative expertise of human resource practitioners on the job performance: Mediating role of achievement motivation," *International Journal of Management*, vol. 12, no. 4, pp. 408–421, 2021, doi: 10.34218/IJM.12.4.2021.035.

[63] M. Asif, A. Khan, and M. A. Pasha, "Psychological capital of employees' engagement: Moderating impact of conflict management in the financial sector of Pakistan," *Global Social Sciences Review*, vol. 4, no. 3, pp. 160–172, 2019, doi: 10.31703/gssr.2019(IV-III).15.

[64] M. A. Pasha, M. Ramzan, and M. Asif, "Impact of economic value-added dynamics on stock prices fact or fallacy: New evidence from nested panel analysis," *Global Social Sciences Review*, vol. 4, no. 3, pp. 135–147, 2019, doi: 10.31703/gssr.2019(IV-III).13.

Author Biographies

Akib Rahman, Master of Information Systems Technologies (Information Assurance and Web Design), Wilmington University, New Castle, Delaware, USA

Master of Information Systems Technologies (Information Assurance and Web Design),

Wilmington University, New Castle, Delaware, USA

Email: akibrahman.edu@gmail.com

Sharmin Sultana, Master of Information Systems Technologies (Information Assurance and Web Design), Wilmington University, New Castle, Delaware, USA

Master of Information Systems Technologies (Information Assurance and Web Design),

Wilmington University, New Castle, Delaware, USA



Published

31-12-2023

Issue

[Vol. 1 No. 1 \(2023\): Journal of Engineering and Computational Intelligence Review](#)

License

Copyright (c) 2023 Akib Rahman, Sharmin Sultana (Author)



This work is licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](https://creativecommons.org/licenses/by-nc-sa/4.0/).

How to Cite

Real-Time Threat Intelligence Correlation and Triage for Reducing Security Analyst Burnout. (2023). *Journal of Engineering and Computational Intelligence Review*, 1(1), 64-86. <https://jecir.com/index.php/jecir/article/view/36>

More Citation Formats

Share



Similar Articles

- Syed Tanvirul Hasan, [Machine Learning Models for Forecasting Employee Demand in Healthcare HR](#), [Journal of Engineering and Computational Intelligence Review: Vol. 3 No. 2 \(2025\): Journal of Engineering and Computational Intelligence Review](#)
- Abdul Hanan, Abdul Hanan Imtiaz Ahmed Khan, Faez Akhtar, Zerminey Saleem, [Autonomous Threat Detection Using Multi-Agent AI and LLM-Assisted Network Traffic Analysis](#), [Journal of Engineering and Computational Intelligence Review: Vol. 4 No. 1 \(2026\): Journal of Engineering and Computational Intelligence Review](#)
- Syed Mehzeyar Ali Zaidi, Sufyan Muhammad Khan, [AI-Powered Cybersecurity: Graph-Based Anomaly Detection in Network Traffic](#), [Journal of Engineering and Computational Intelligence Review: Vol. 4 No. 1 \(2026\): Journal of Engineering and Computational Intelligence Review](#)
- Muhammad Danish, Malik Muhammad Siraj, [AI and Cybersecurity: Defending Data and Privacy in the Digital Age](#), [Journal of Engineering and Computational Intelligence Review: Vol. 3 No. 1 \(2025\): Journal of Engineering and Computational Intelligence Review](#)

- Umma Twaha, [Mitigating Financial Waste in the U.S. Healthcare System: An AI-Driven Framework for Real-Time Fraud Detection in Medicare and Medicaid Claims](#), [Journal of Engineering and Computational Intelligence Review: Vol. 2 No. 2 \(2024\): Journal of Engineering and Computational Intelligence Review](#)
- Sameer Niazi, [Big Data Analytics with Machine Learning: Challenges, Innovations, and Applications](#), [Journal of Engineering and Computational Intelligence Review: Vol. 2 No. 1 \(2024\): Journal of Engineering and Computational Intelligence Review](#)
- Imran Nazeer Butt, Sameer Shah, [The Convergence of Artificial Intelligence and 6G Networks: A Transformative Shift in Computing and Communications](#), [Journal of Engineering and Computational Intelligence Review: Vol. 3 No. 1 \(2025\): Journal of Engineering and Computational Intelligence Review](#)
- Asma Shaheen, [The Internet of Things \(IoT\): A Comprehensive Review of Technologies, Applications, Challenges, and Future Trends](#), [Journal of Engineering and Computational Intelligence Review: Vol. 2 No. 1 \(2024\): Journal of Engineering and Computational Intelligence Review](#)
- Syed Muhammad Kashif , Fariba Chowdhury, [A Comprehensive Review of Asset Management Systems: Trends, Technologies, and Future Directions](#), [Journal of Engineering and Computational Intelligence Review: Vol. 2 No. 2 \(2024\): Journal of Engineering and Computational Intelligence Review](#)
- Fahad Amin, Ihsan Said, Mahad Ali Butt, [AI-Based Cybersecurity Solutions: Securing Information and Privacy in the Evolving Digital Age](#), [Journal of Engineering and Computational Intelligence Review: Vol. 2 No. 2 \(2024\): Journal of Engineering and Computational Intelligence Review](#)

1-10 of 50 Next →

You may also [start an advanced similarity search](#) for this article.

For online submission

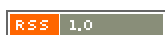
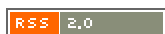
Please send your manuscript/ research articles on one of the following Email Address:

1. support@jecir.com
2. jecir365@gmail.com

[Make a Submission](#)

Latest publications

ATOM 1.0



Information

[For Readers](#)

[For Authors](#)

[For Librarians](#)

Browse

[Categories](#)

Language

[English](#)

[Open Journal Systems](#)

Keywords

**Journal of Engineering and Computational Intelligence Review, ISSN (Online): 3105-4811
ISSN (Print): 3106-969X**

EMAIL: support@jecir.com

Contact & WhatsApp No. +92-330-1879000

This work is licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](#).



[More information
about the](#)

